Basic Traffic Models and Traffic Waves

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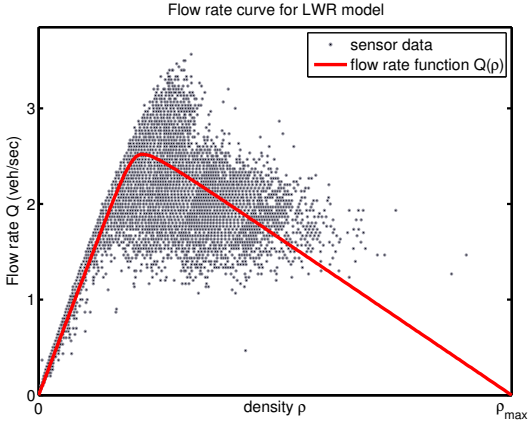
Fundamental Diagram



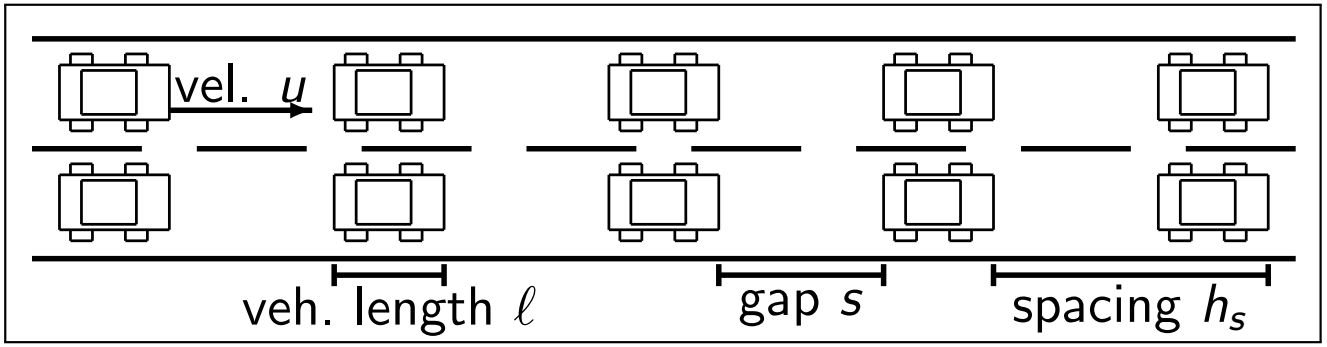
Ring Road

September 16–17, 2020

Mathematical Challenges and Opportunities for Autonomous Vehicles Tutorials

Institute for Pure and Applied Mathematics, UCLA

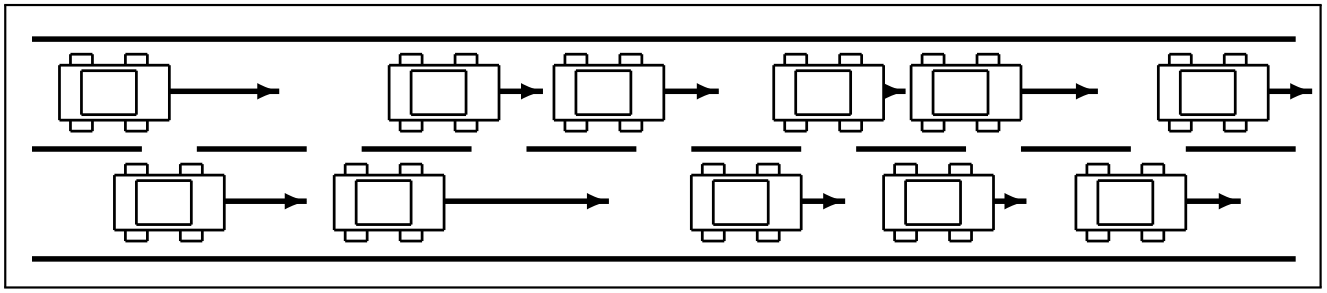
Uniform traffic flow



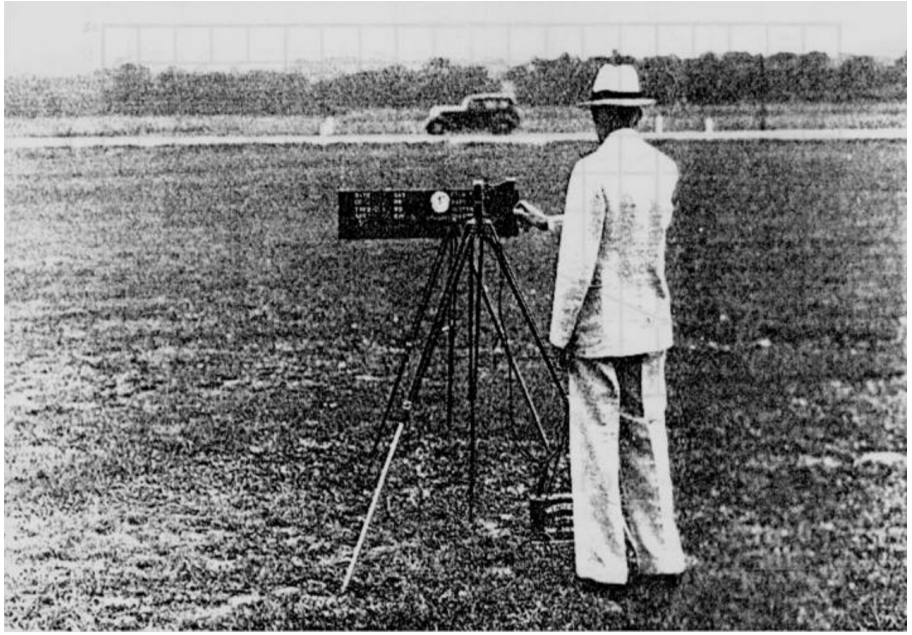
Fundamental quantities

- density ρ: # vehicles per unit length; $\rho_{\max}$: bumper to bumper + safety
- flow rate (throughput) q: # vehicles passing fixed position per time
- velocity u: distance traveled per unit time
- bulk-velocity $u = q/\rho$: correct notion in non-uniform flow
- spacing h_s: road length per vehicle; gap $s = h_s - \ell$

Non-uniform traffic flow

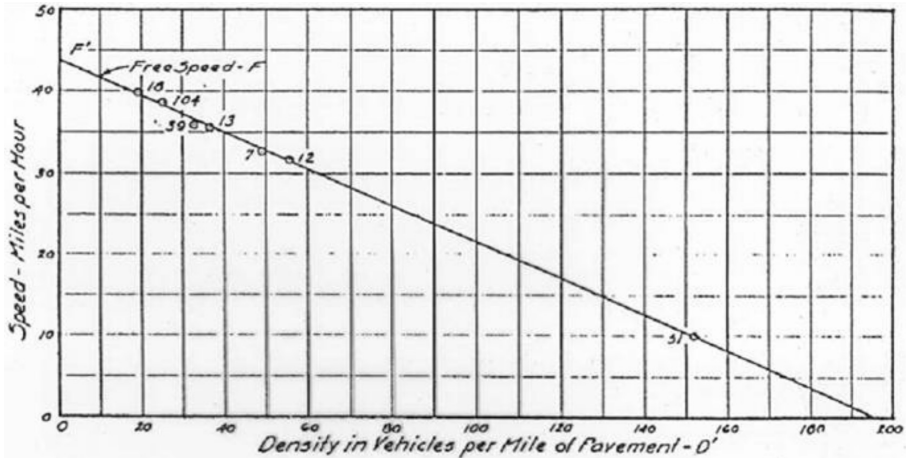


Bruce Greenshields collecting data (1933)



[This was only 25 years after the first Ford Model T (1908)]

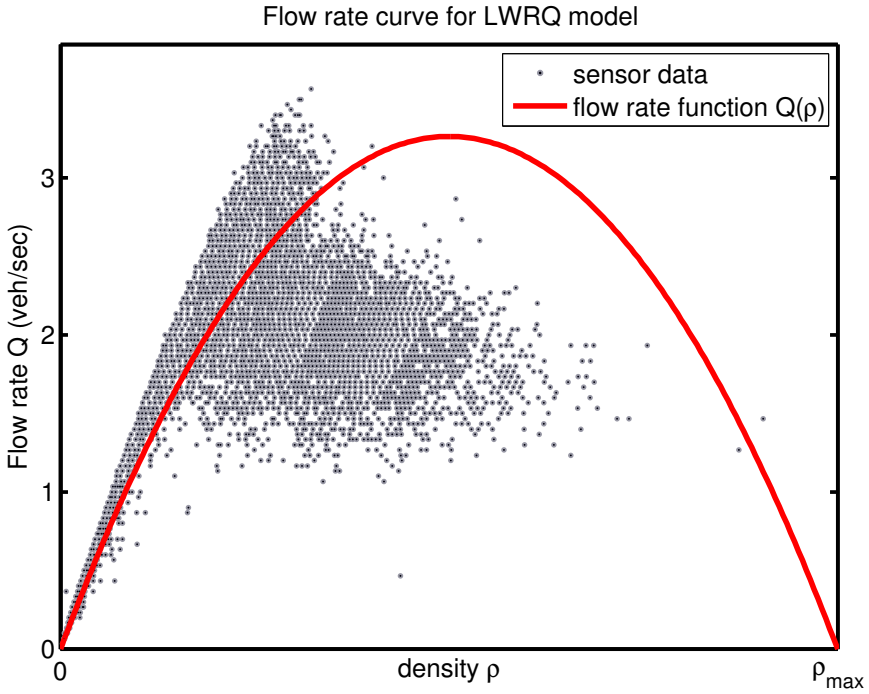
Postulated density–velocity relationship



Deduced relationship

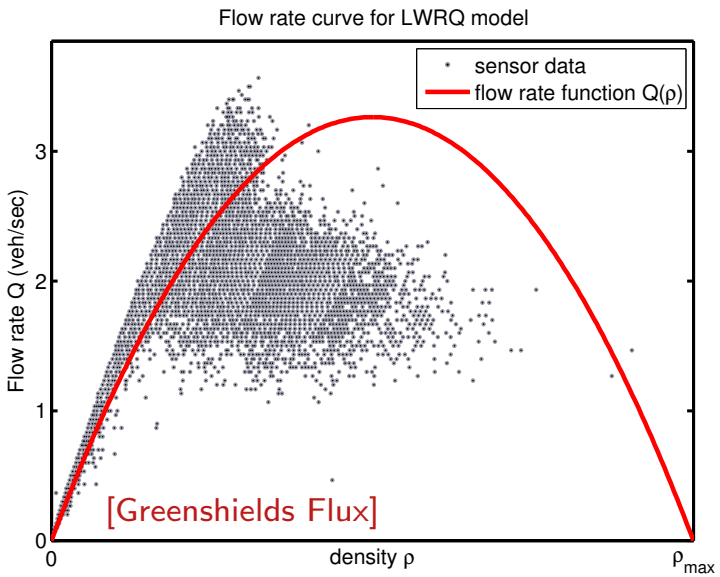
- $u = U(\rho) = u_{\max}(1 - \rho/\rho_{\max})$,
  $\rho_{\max} \approx 195$ veh/mi; $u_{\max} \approx 43$ mi/h
- Flow rate
  $q = Q(\rho) = u_{\max}(\rho - \rho^2/\rho_{\max})$

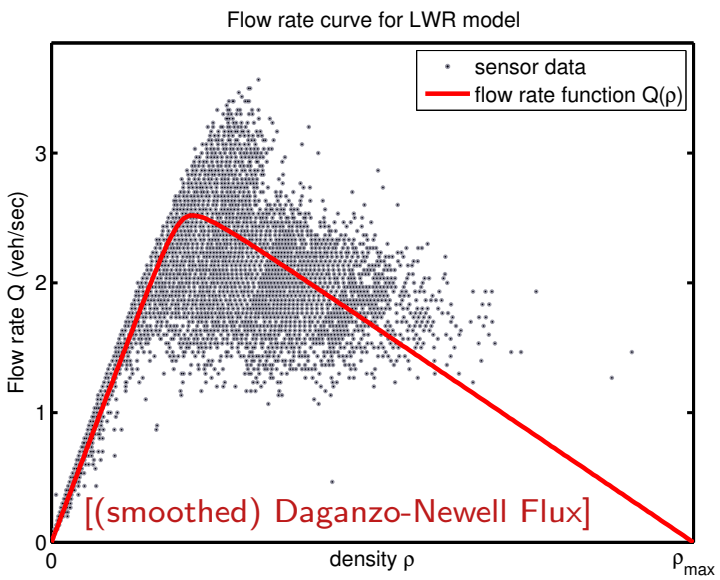
Contemporary measurements (q vs. ρ)



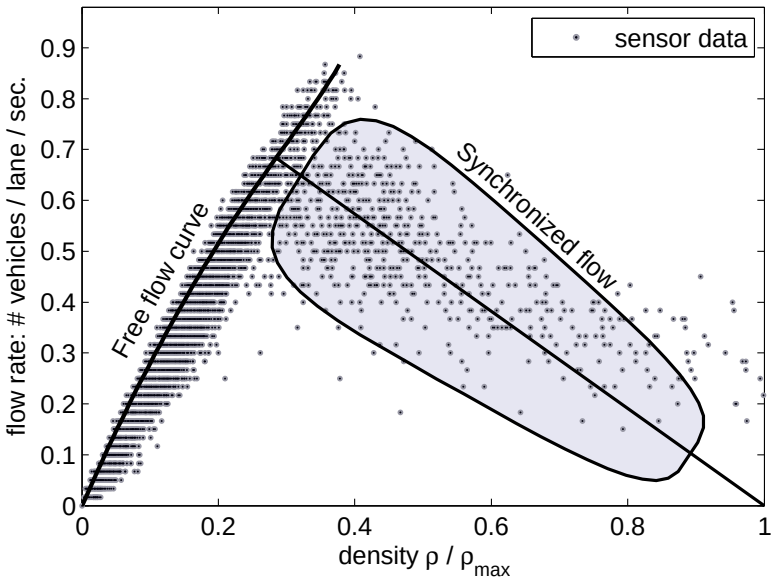
[Fundamental Diagram of Traffic Flow]

Fundamental Diagram (FD) of traffic flow (detector data)





Traffic phase theory (here: 2 phases) [Kerner]



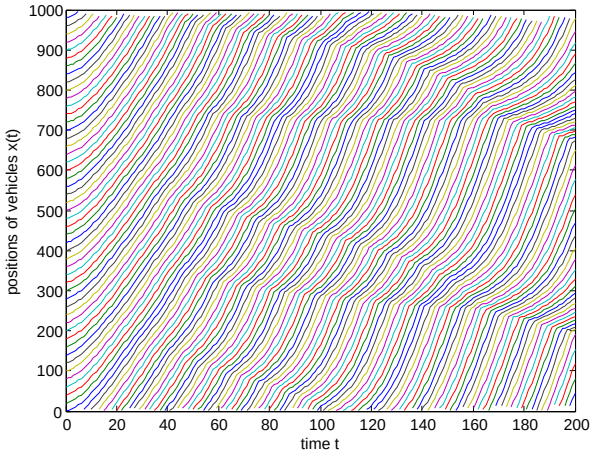
FDs around the world exhibit same features

- for ρ small (free-flow): small spread
- above a critical density (congestion): $Q(\rho)$ decreasing & FD set-valued

Key open question in traffic flow theory: precise phenomenological understanding of spread (role of sensor noise, inhomogeneities, non-equilibrium effects, etc.).

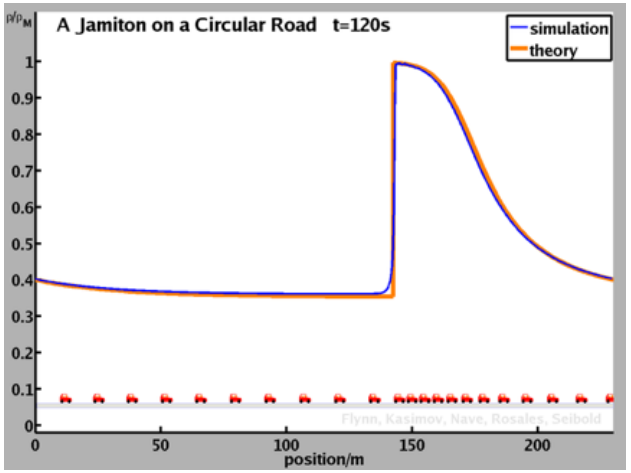
Microscopic Models

$$\ddot{x}_j = G(x_{j+1} - x_j, u_j, u_{j+1})$$

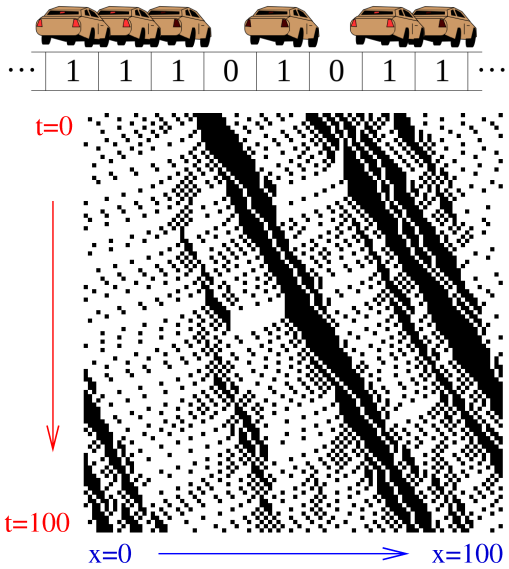


Macroscopic Models

$$\begin{cases} \rho_t + (\rho u)_x = 0 \\ (u+h)_t + u(u+h)_x = \frac{1}{\tau}(U-u) \end{cases}$$



Cellular Models



Idea

Describe behavior of individual vehicles (ODE system).

Micro $\longleftrightarrow$ Macro

- macro = limit of micro when #vehicles $\to \infty$
- micro = discretization of macro in Lagrangian variables

Methodology and role

- Describe aggregate/bulk quantities via PDE.
- Natural framework for multiscale phenomena, traveling waves, and shocks.
- Suitable framework to incorporate sparse data [Mobile Millennium Project].

Idea

Cell-to-cell propagation (space-time-discrete).

Cellular $\longleftrightarrow$ Macro

- macro = limit of cellular
- cellular = discretization of macro in Eulerian variables

Key Distinction for All Traffic Models

- **First-order dynamics:** System state is vehicle positions (or density). Obtain (instantaneous) vehicle velocities from positions.
- **Second-order dynamics:** System state is vehicle positions and velocities. Model vehicle accelerations (Newton's laws of motion).

- First-order dynamics can produce shock waves (moving upstream end of traffic jam; red/green light dynamics); but ...
- Second-order dynamics needed to produce instabilities and traveling waves (phantom traffic jams). [Or: first-order with delay; not treated here]

Microscopic Models

First-order: $\dot{x}_j = F(x_{j+1} - x_j)$

Second-order:

$$\ddot{x}_j = G(x_{j+1} - x_j, u_j, u_{j+1})$$

Macroscopic Models

First-order: $\rho_t + (\rho U(\rho))_x = 0$

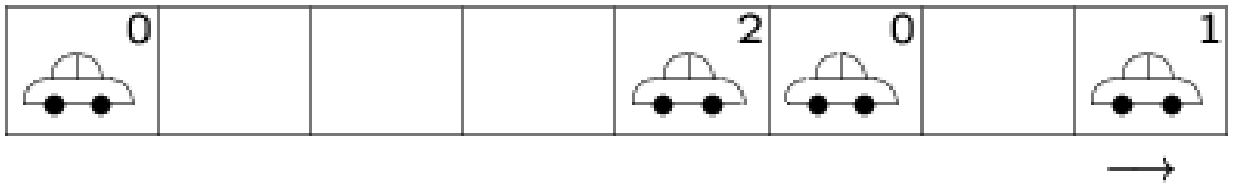
Second-order:

$$\begin{cases} \rho_t + (\rho u)_x = 0 \\ (u + h(\rho))_t + u(u + h(\rho))_x = \frac{1}{\tau}(U(\rho) - u) \end{cases}$$

Traffic models

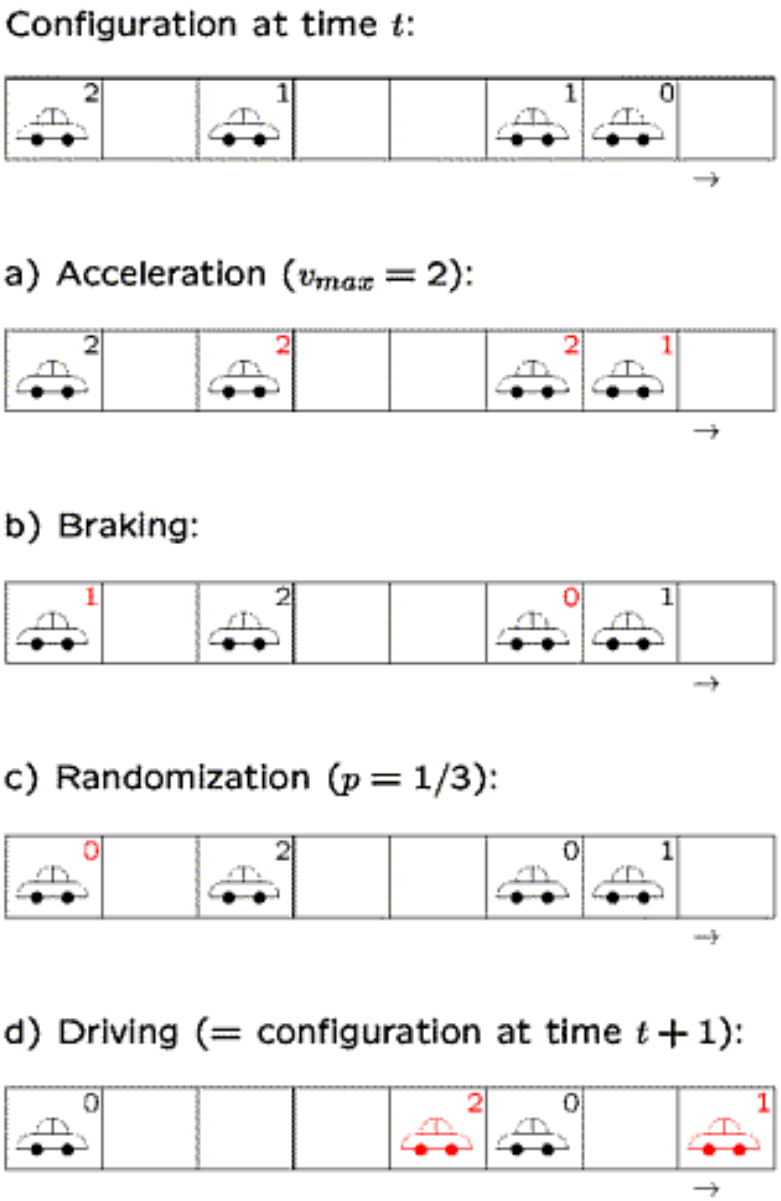
Nagel–Schreckenberg model

- Periodic 1d lattice (ring) Autobahn
- discretized in space and time
- Cars occupying a lattice moving with velocities $v = 0, 1, 2, 3, 4, 5$
- Remark, if max speed is 126 km/h, then lattice length is 7 m, a very good guess for a car in a traffic jam
- It uses parallel update: at each timestep all cars move v sites forward

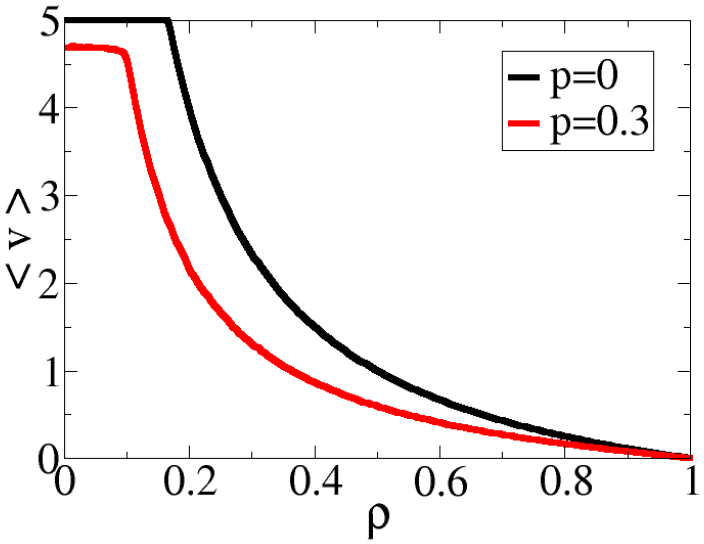


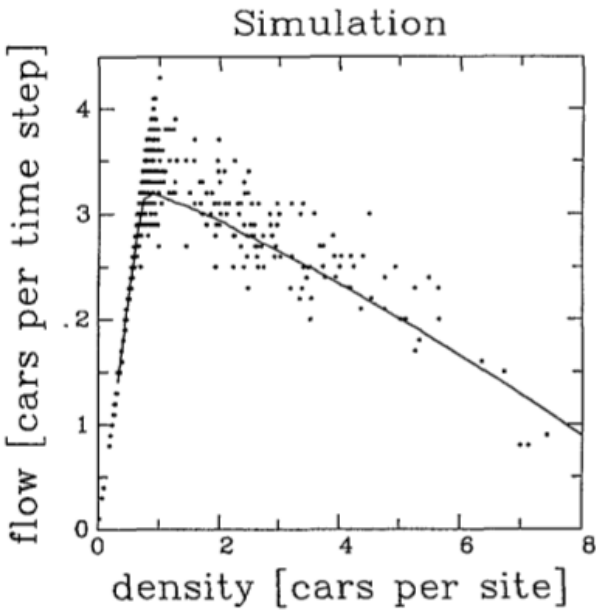
Nagel–Schreckenberg model

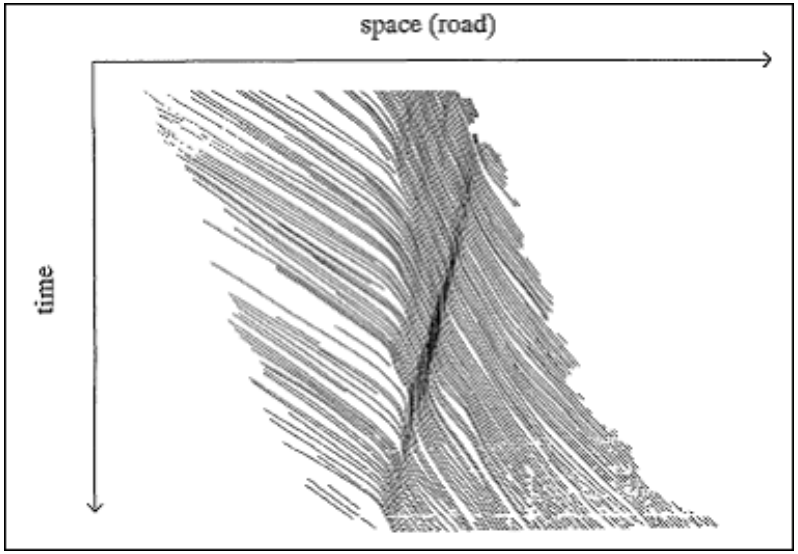
- Algorithm:
  1. **Acceleration**: All cars not at the maximum velocity increase their velocity by 1
  2. **Slowing down**: Speed is reduced to distance ahead (1 sec rule)
  3. **Randomization**: With probability p speed is reduced by 1
  4. **Car motion**: Each car moves forward the number of cells equal to their velocity.

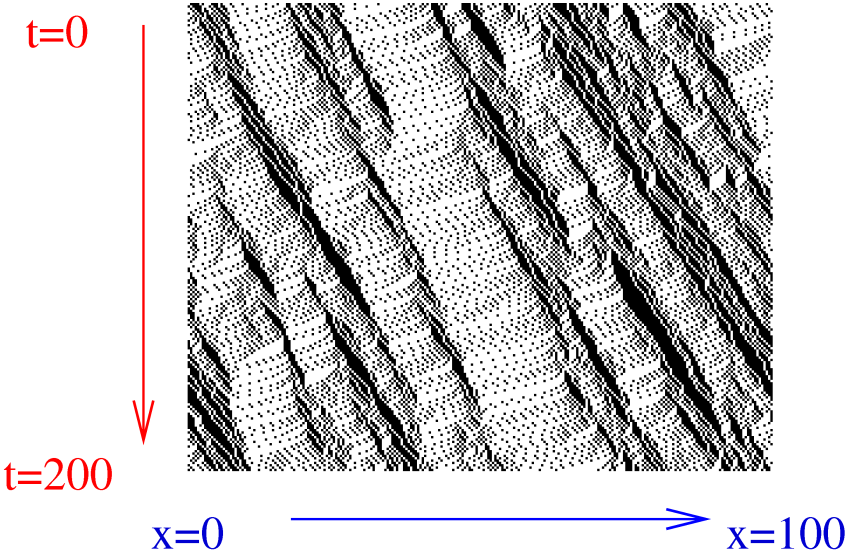


Emergence of traffic jams



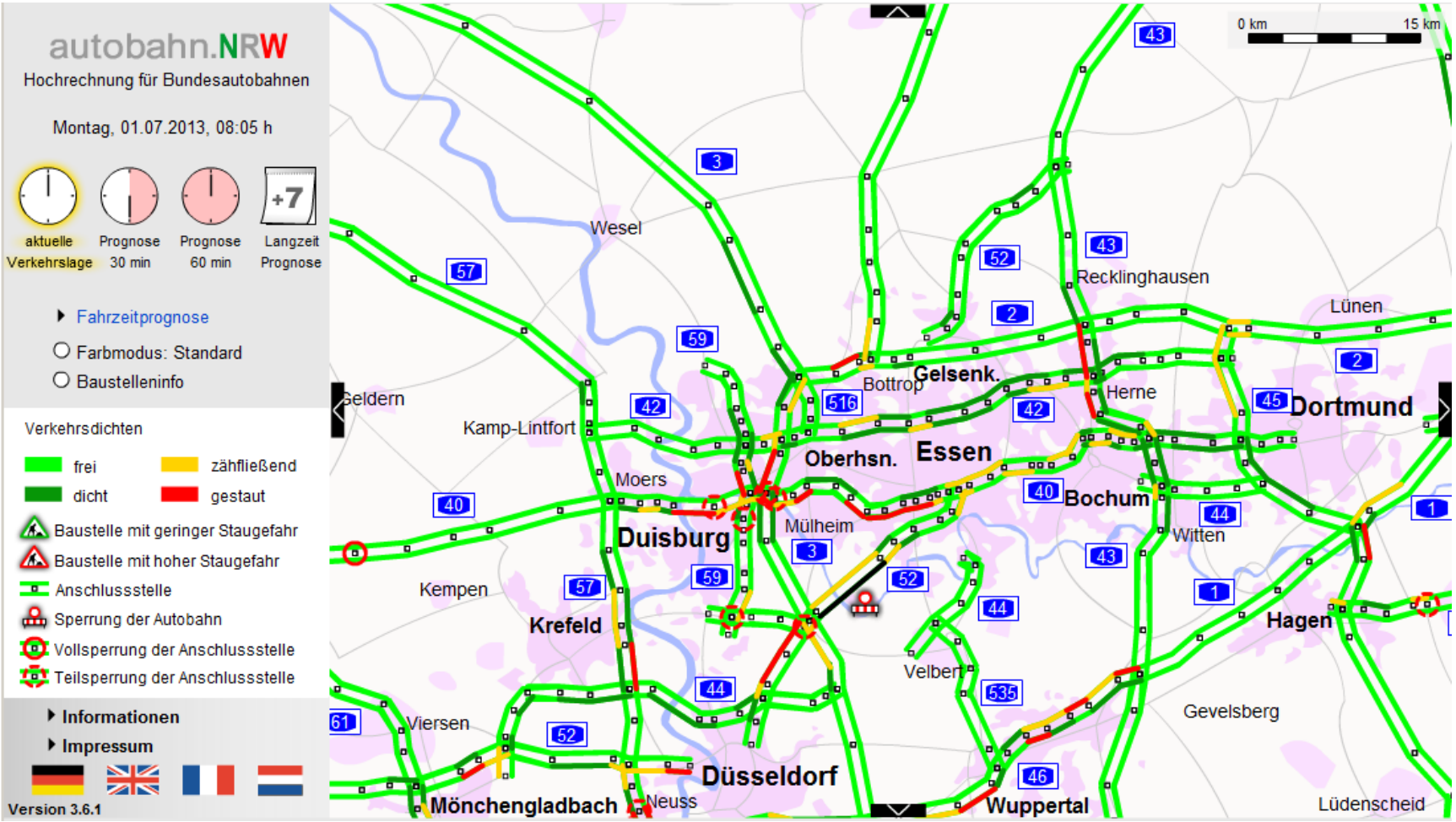






Nagel–Schreckenberg model

- Transition from free-flow to jammed state
- Jammed state is a phase-separated phase
- Without randomization a sharp transition
- Had been used in NRW to predict traffic jams


autobahn.NRW
Hochrechnung für Bundesautobahnen
Montag, 01.07.2013, 08:05 h
aktuelle Verkehrslage
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Prognose 60 min
Langzeit Prognose
Fahrzeitprognose
Farbmodus: Standard
Baustelleninfo
Verkehrsdichten
frei
zähfließend
dicht
gestaut
Baustelle mit geringer Staugefahr
Baustelle mit hoher Staugefahr
Anschlussstelle
Sperrung der Autobahn
Vollsperrung der Anschlussstelle
Teilsperrung der Anschlussstelle
Informationen
Impressum
Version 3.6.1
0 km
15 km
Wesel
Recklinghausen
Lünen
Gelsenk.
Bottrop
Herne
Dortmund
Kamp-Lintfort
Oberhsn.
Essen
Moers
Bochum
Mülheim
Duisburg
Witten
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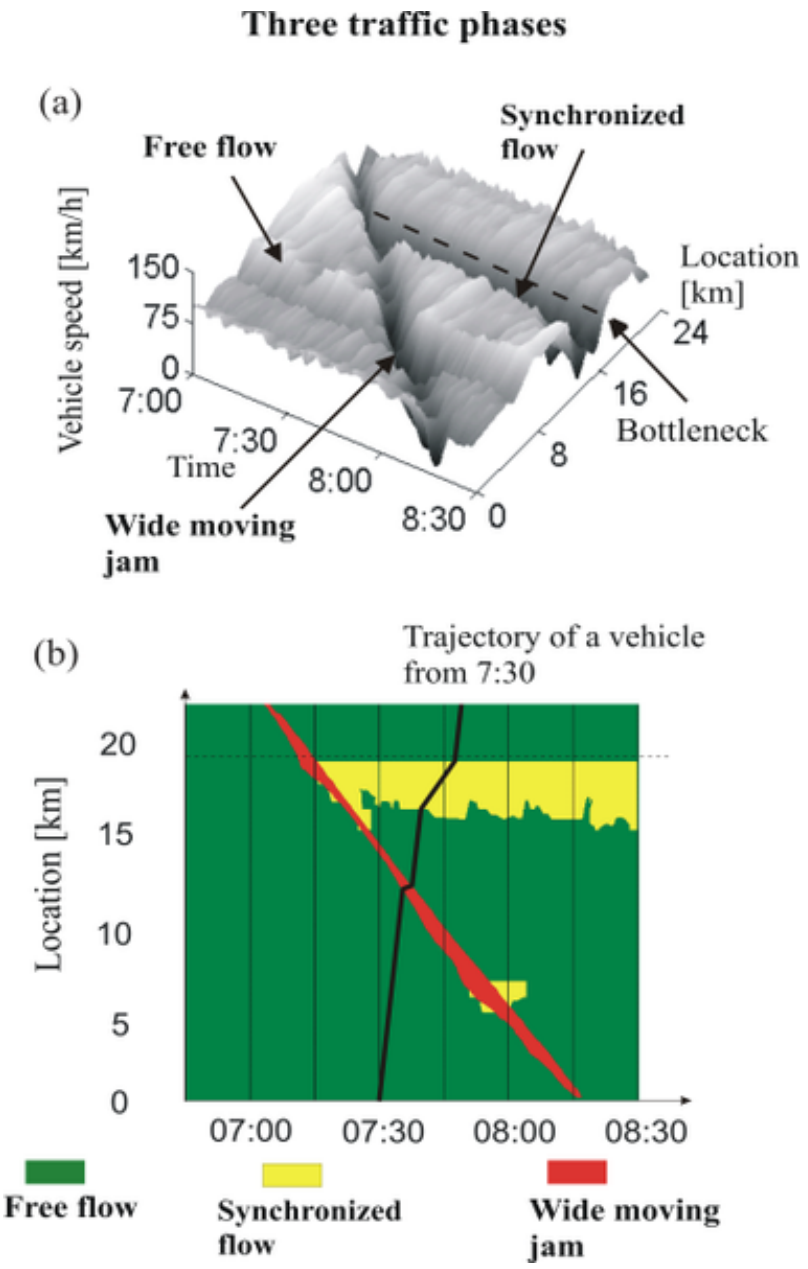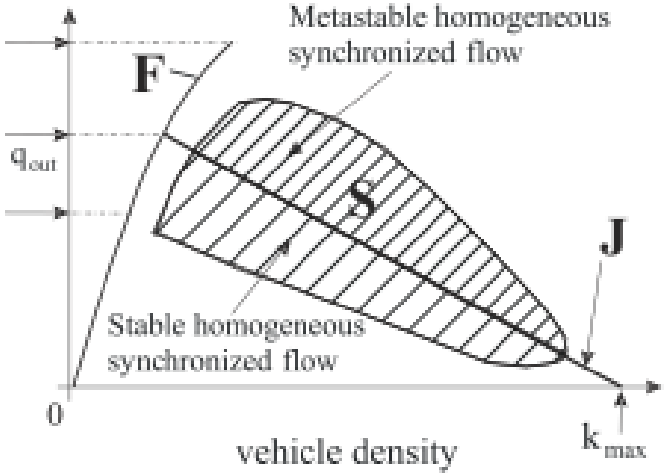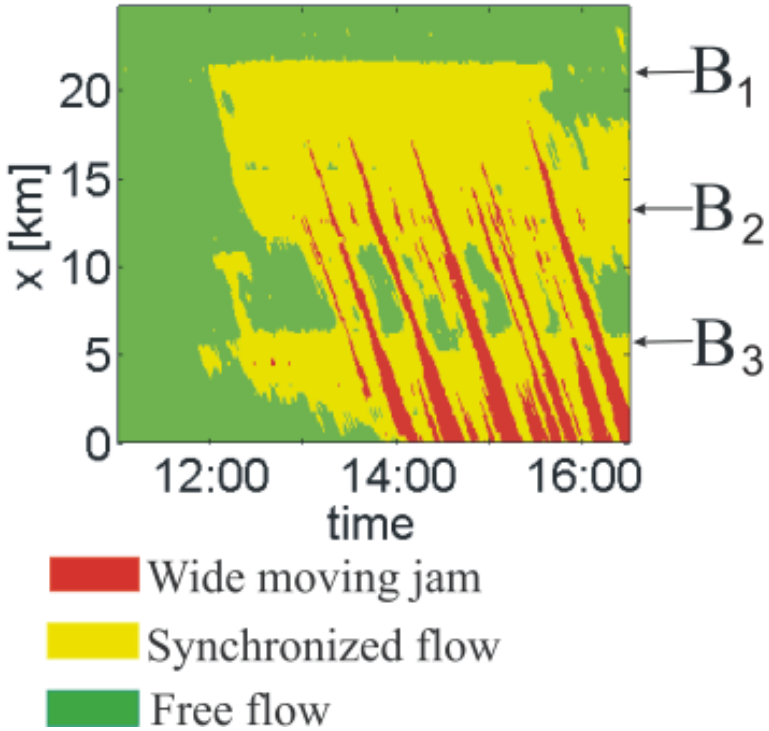
Three-phase traffic theory